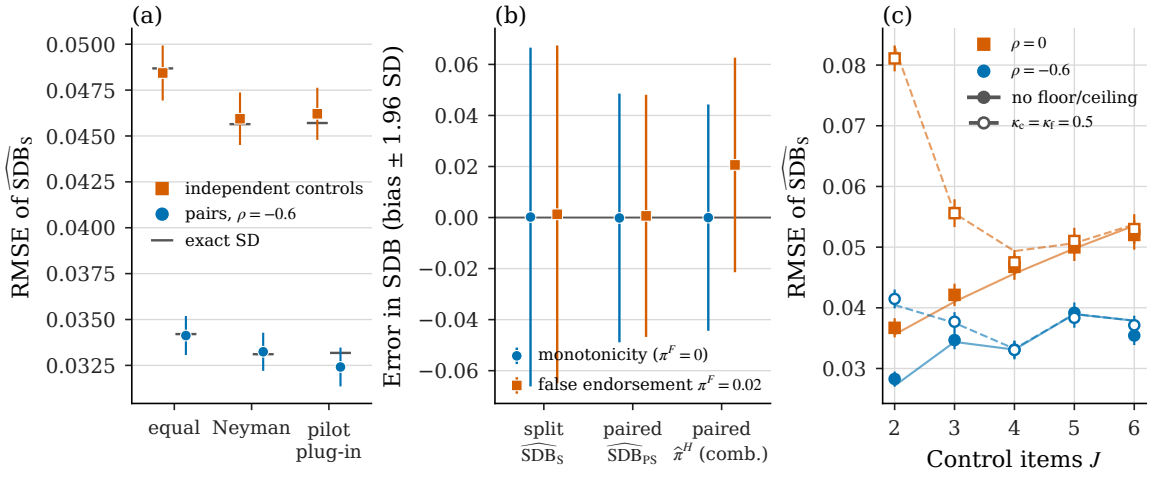
(a)
RMSE of $\widehat{\mathrm{SDB}}_S$
0.0500
0.0475
0.0450
0.0425
0.0400
0.0375
0.0350
0.0325
independent controls
pairs, $\rho = -0.6$
exact SD
equal
Neyman
pilot plug-in
(b)
Error in SDB (bias ± 1.96 SD)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
monotonicity ($\pi^F = 0$)
false endorsement $\pi^F = 0.02$
split $\widehat{\mathrm{SDB}}_S$
paired $\widehat{\mathrm{SDB}}_{PS}$
paired $\hat{\pi}^H$ (comb.)
(c)
RMSE of $\widehat{\mathrm{SDB}}_S$
0.08
0.07
0.06
0.05
0.04
0.03
$\rho = 0$
$\rho = -0.6$
no floor/ceiling
$\kappa_c = \kappa_t = 0.5$
2
3
4
5
6
Control items J